Oppgaver for group sessions uke 6 (02.02-06.02)

MNF130

Spring 2026

Oppgavene er hentet fra læreboken K. Rosen: Discrete Mathematics and its Applications, kapittel 1.4.

1. Let $P(x)$ be the statement "The word x contains the letter a". What are these truth values?
   (a) $P(\text{orange})$.
   (b) $P(\text{lemon})$.
   (c) $P(\text{true})$.
   (d) $P(\text{false})$.

2. Translate these statements into English, where $C(x)$ is "x is a comedian" and $F(x)$ is "x is funny" and the domain consists of all people.
   (a) $\forall x(C(x) \to F(x))$.
   (b) $\forall x(C(x) \land F(x))$.
   (c) $\exists x(C(x) \to F(x))$.
   (d) $\exists x(C(x) \land F(x))$.

3. Let $P(x)$ be the statement "x can speak Russian" and let $Q(x)$ be the statement "x knows the computer language C++". Express each of these sentences in terms of $P(x)$, $Q(x)$, quantifiers, and logical connectives. The domain for quantifiers consists of all students at your school.
   (a) There is a student at your school who can speak Russian and who knows C++.
   (b) There is a student at your school who can speak Russian but who doesn't know C++.
   (c) Every student at your school either can speak Russian or knows C++.
   (d) No student at your school can speak Russian or knows C++.

4. Determine the truth value of each of these statements if the domain for all variables consists of all integers.
   (a) $\forall n(n^2 \geq 0)$.

(b) $\exists n(n^2 = 2)$.

(c) $\forall n(n^2 \geq n)$.

(d) $\exists n(n^2 < 0)$.

5. Express the negation of each of these statements in terms of quantifiers without using the negation symbol.

   (a) $\forall x(x > 1)$.

   (b) $\forall x(x \leq 2)$.

   (c) $\exists x(x \geq 4)$.

   (d) $\exists x(x < 0)$.

   (e) $\forall x((x < -1) \vee (x > 2))$.

   (f) $\exists x((x < 4) \vee (x > 7))$.

6. Translate these specifications into English, where $F(p)$ is "Printer p is out of service", $B(p)$ is "Printer p is busy", $L(j)$ is "Print job j is lost" and $Q(j)$ is "Print job j is queued".

   (a) $\exists p(F(p) \wedge B(p)) \rightarrow \exists j L(j)$.

   (b) $\forall p B(p) \rightarrow \exists j Q(j)$.

   (c) $\exists j(Q(j) \wedge L(j)) \rightarrow \exists p F(p)$.

   (d) $(\forall p B(p) \wedge \forall j Q(j)) \rightarrow \exists j L(j)$.

7. As mentioned in the text, the notation $\exists! x P(x)$ denotes *"There exists a unique x such that $P(x)$ is true"*. If the domain consists of all integers, what are the truth values of these statements?

   (a) $\exists! x(x > 1)$.

   (b) $\exists! x(x^2 = 1)$.

   (c) $\exists! x(x + 3 = 2x)$.

   (d) $\exists! x(x = x + 1)$.

8. Let $P(x)$, $Q(x)$ and $R(x)$ be the statements "x is a professor", "x is ignorant" and "x is vain", respectively. Express each of these statements using quantifiers, logical connectives, and $P(x)$, $Q(x)$ and $R(x)$, where the domain consists of all people.

   (a) No professors are ignorant.

   (b) All ignorant people are vain.

   (c) No professors are vain.

   Does (c) follow from (a) and (b)?